Enhancing Employability Through an Advanced Mock Interview Platform for Fresh IT Graduates

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2024 7th International Seminar on Research of Information Technology and Intelligent Systems (ISRITI) | 979-8-3315-1964-3/24/$31.00 ©2024 IEEE | DOI: 10.1109/ISRITI64779.2024.10963436

Abstract— **In today's competitive job market, securing employment depends not only on possessing necessary qualifications but also on demonstrating an exceptional interview performance. Therefore effective preparation for interviews has become crucial for enhancing employability. Traditional mock interviews are often impaired by the lack of personalised feedback and real-time evaluation.This critical gap affects the candidates' ability to identify and address weaknesses, hindering efforts to refine their interviewing skills and maximise their chances of success. This research proposes the development of an advanced AI driven mock interview platform. This platform aims to bridge the existing gap in interview preparation methods by providing candidates with a personalised, interactive, and data-driven experience that empowers them to build confidence. The platform leverages Large Language Models (LLMs) to generate personalised interview questions tailored to a candidates experiences and qualifications. Then utilises Convolutional Neural Networks (CNNs) for multimodal analysis of non-verbal cues, offering insights to a candidate's confidence level. Additionally, it provides real-time feedback on communication skills, focusing on delivery, clarity, and tone. To further personalise the learning experience personalised recommendations will be facilitated. This multifaceted approach seeks to fundamentally redefine interview preparation and equip individuals with the necessary tools and confidence to thrive in the competitive job market.**

Keywords*— Mock Interview Platform, LLM, Personalised Feedback, Interview Preparation, Multimodal Analysis, Real-Time Evaluation, Employability Enhancement, Natural Language Processing***

I. Introduction

Today's job market demands a broader definition of employability, encompassing not just academic qualifications but a diverse skill set and personal attributes. For Information Technology (IT) undergraduates, navigating this landscape presents unique challenges amidst constant technological advancements. One of the primary challenges faced by IT undergraduates lies in the dynamic nature of the IT industry itself. Technological advancements occur at a rapid pace, resulting in constant shifts in skill requirements and industry demands. Consequently, IT undergraduates often find themselves dealing with the need to continually update their skill sets to remain relevant in the job market. Additionally, the expansion of IT-related courses has led to a saturation of the talent pool, intensifying competition for sought employment opportunities.

Traditional methods of securing employment, such as job interviews, pose significant hurdles for IT undergraduates. While possessing technical expertise is crucial, many employers also seek candidates with well-developed soft skills, including communication, problem-solving, and teamwork. However, IT undergraduates may struggle to effectively communicate their technical knowledge during interviews, thereby limiting their employability prospects.

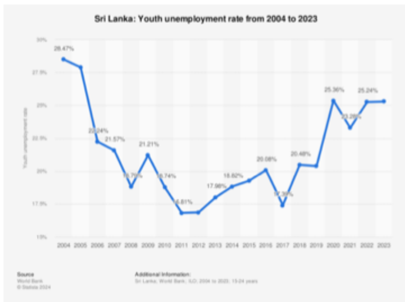

Fig 1: Sri Lanka Youth Unemployment Rate from 2004 to 2023

This research explores an approach to enhance the employability of IT undergraduates by addressing the gaps in traditional interview preparation methods. By developing an advanced mock interview platform driven by artificial intelligence (AI), this research aims to provide IT undergraduates with personalised feedback, real-time assessment, and tailored learning recommendations. The integration of technologies such as natural language processing and machine learning enables the generation of adaptive interview questions tailored to individual candidate profiles. Additionally, multimodal analysis techniques, including facial recognition and verbal analysis, offer a comprehensive evaluation of candidates' skills and abilities. By empowering IT undergraduates with the tools and resources necessary to navigate the complexities of the job market, this research seeks to redefine interview preparation paradigms and enhance their employability prospects. By addressing the challenges faced by IT undergraduates and offering innovative solutions, this research endeavours to equip them with the skills and confidence needed to succeed in securing desirable employment opportunities

II. LITERATURE SURVEY

In terms of question generation, current interview question generation systems suffer from significant deficiencies, including reliance on static question pools that fail to adapt to industry dynamics[1]. This results in a lack of diversity and relevance, hindering the assessment process. Additionally, these systems often draw from fixed sources, leading to outdated questions that do not reflect the latest industry developments [2]. Personalisation is another issue, as these systems neglect to consider candidates' unique experiences and qualifications, resulting in questions that may not effectively assess suitability for specific roles [3]. According to a study conducted by Y.Miao and W.Huag on the Interaction Design of Mock interview application, existing systems lack diversity and relevance to candidates' backgrounds and job requirements[4]. Many systems solely rely on predefined question sets that do not consider individual candidate resumes or specific job roles [1]. This lack of customisation leads to questions that may not effectively assess a candidate's suitability for the position. Addressing these challenges requires dynamic approaches that incorporate machine learning techniques and feedback mechanisms[5].

The employability of individuals using communication skills has been given much importance in numerous studies. Robbins et al. [6] emphatically emphasised that for effective teamwork and client interactions, good communication is required in the IT industry. Several works have also studied how technology can be integrated to improve communication skills. For instance, Chandrasekaran et al. [7] recently showed that speech-to-text technologies could significantly enhance language learning and be useful for language proficiency improvements. Brown et al. [8] hinted at some of the potentials in LLMs, such as GPT-3, for language understanding and grammar correction, which provided a base for our approach. The recent advances in AI and NLP within the last couple of years set the ground for the development of very robust tools for communication analysis. The first pre-trained LLMs started to show that, in the case of their adaptation for speech transcription (for example, Whisper), even further fine-tuning could support real-time implementations. Fine-tuning LLMs for a particular task has proved successful, especially for educational and professional training purposes. To elaborate on the work done previously, this paper merges those findings with those technologies into a mock interview platform that will build IT graduates' communication skills.

Confidence refers to trust and a strong belief in something or someone. During an interview, demonstrating self-confidence—showing trust and faith in your own abilities—can be the deciding factor for your success compared to other candidates [10]. Deep learning techniques, especially Convolutional Neural Networks (CNNs), are effective for image classification tasks involving confidence detection. Research using a deep learning-based approach to measuring confidence in virtual interviews shows that CNN models can classify facial expressions to measure self-confidence and provide valuable feedback to help interviewers improve [10].

The use of deep learning techniques in the field of facial behaviour analysis has seen significant growth, particularly in the context of enhancing interview performance. And in recent years, interest in automatic analysis and understanding of facial behaviour has been increasing [10]. Confidence, a critical factor in job interviews, is often evaluated through non-verbal cues such as facial expressions, body language, and vocal tone[11]. Despite humans recognising facial expressions effortlessly, developing reliable machine-based expression recognition systems remains a challenge.Recent advancements in face detection, feature extraction mechanisms, and expression classification techniques have been notable, yet creating an automated system for accurate facial expression recognition is complex [9].Recent advances in computer vision and machine learning are creating enough technology to develop sophisticated tools that analyse these cues to provide a comprehensive assessment of a candidate's confidence. Research has demonstrated the effectiveness of CNN in facial expression classification and confidence level detection. CNNs can classify facial expressions such as anger, happiness, fear, sadness, disgust, and neutrality from images[9].One notable contribution in this field is OpenFace, an open-source tool designed for researchers and developers. OpenFace excels in multiple facial behaviour analysis tasks, including facial landmark detection, head pose estimation, facial action unit recognition, and eye-gaze estimation. Its state-of-the-art computer vision algorithms enable real-time performance using a simple webcam, making it accessible and efficient without requiring specialised hardware. Moreover, OpenFace's lightweight messaging system facilitates easy integration with other applications, enhancing its utility for building interactive systems based on facial behaviour analysis [10].

III. METHODOLOGY

The methodology encompasses a systematic approach to realising the proposed solution, consisting of distinct phases to enhance interview preparation comprehensively.

First the user is allowed to upload their Resume and Job Description to the platform, initiating the screening process. Subsequently, the system analyses the candidate's expertise and target job position, generating a score for the reuse and Job description similarity and then generating a tailored question pool for the mock interview experience. A multi-modal approach is then employed to gauge the candidate's confidence levels during the mock interview. Facial recognition technology is integrated to assess emotions, eye movements, and body language. An emotion detection system provides feedback on confidence levels, while machine learning algorithms identify patterns in facial expressions correlated with confidence. Next, the system focuses on refining candidates' communication skills through an Adaptive Linguistic Feedback Engine. Interview transcripts undergo analysis for grammar checks and identification of communication gaps. Context-aware suggestions are provided for improvement, along with tone and formality detection to tailor recommendations to the professional context. Finally, a comprehensive evaluation of the candidate's performance is conducted, identifying strengths and weaknesses in technical and soft skills. Feedback is provided, along with suggestions for related job vacancies and skill development courses.

This methodology offers a robust and comprehensive framework for enhancing interview preparation in today's competitive job market.

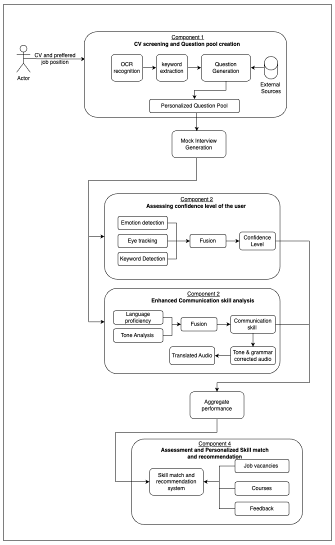

Fig 2: System Overview

A. Optimising Mock Interview Question Generation

This module , takes in reuse and job description input and aims to identify the similarity and consequently generating high-quality interview questions tailored to specific skills of a candidate.

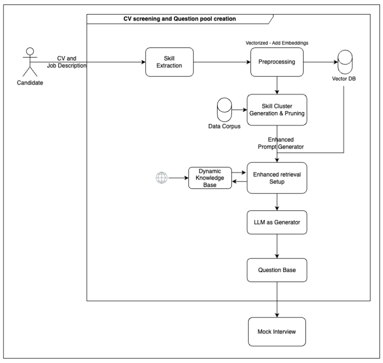

Fig 3: Question Generation Module Architecture

1) Resume and job description similarity scoring

An input module facilitates resume and job description ingestion, offering file uploads or direct text input. The Resume and JD similarity analysis begins with information extraction using Spacy's PhraseMatcher to identify entities from job descriptions. A dictionary containing categories for education, majors, and skills is utilised for entity recognition. Matching rules are then applied to assess similarities between resumes and job descriptions, incorporating both keyword and ontology-based techniques. For skill matching, semantic similarity is evaluated using various pre-trained models to generate word embeddings, which are compared using cosine similarity. This approach ensures a comprehensive comparison by embedding semantic meaning into vectors and calculating similarity between resume and job description vectors.

2) Skill cluster generation and identification of latent skills of candidates

Initially, the system uses information extraction techniques to identify and categorise skills from resumes. This step involves applying entity recognition to extract relevant skills and attributes.Once skills are extracted, the system employs clustering algorithms to group similar skills and identify latent skills. Specifically, KMeans is used to cluster the extracted skills into distinct groups. The algorithm is first applied to a pre-processed dataset to determine the optimal number of clusters, based on minimising clustering errors.To provide a more nuanced skill assessment, the augmented skills are extracted from the two clusters nearest to the candidate's predicted cluster. This involves calculating distances between clusters to determine proximity, and then selecting skills from the top clusters closest to the candidate's cluster. This approach ensures that the skill recommendations are both relevant and comprehensive, capturing both explicit and latent skills.

3) Tailored Question Generation

These structured skill clusters then inform the generation of interview questions, ensuring they are targeted and comprehensive. Content according to the skills are retrieved from a dynamic data corpus, which includes web-crawled data from sources such as W3Schools. This dynamic corpus provides up-to-date and relevant content related to the extracted skills. The retrieved content is then tokenised using <h> tag to prepare it for input into the LLM. This tokenised content serves as an additional prompt, enhancing the model's ability to generate questions that are highly relevant to the specific skills and current industry standards. Key tools such as PyTorch and Transformers facilitate model training and deployment, while Hugging Face's pipelines streamline the integration of state-of-the-art models like T5 for text generation tasks. By integrating dynamically sourced data with the LLM, the questions generated are more precise and reflective of the latest industry practices.

B. Speech Empowerment Through Intelligent Verbal Analysis

Analysis within The Enhanced Communication Skills Component is multi-step and devised to provide users with articulate feedback for their communication abilities specifically. The processes are described below.

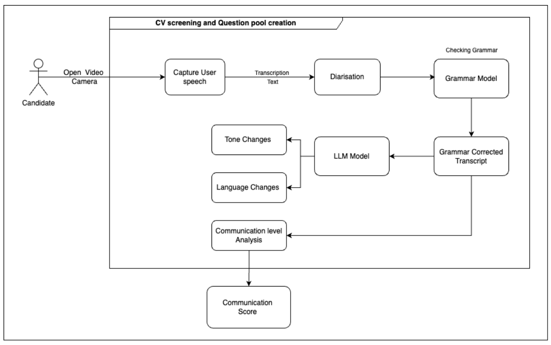

Fig 4: Speech Empowerment Module Architecture

1) *Speech-to-Text Transcription*

During the mock interview, the user's speech is captured and transcribed using Whisper, a pre-trained model for speech-to-text. Whisper was selected for its accuracy and adaptability in working with different accents and speech patterns. This process guarantees that the transcription faithfully reflects the user's spoken words.

2) Grammatical Error Corrections

The text is run through the LLM first, fine-tuned for grammatical error corrections based on the type of errors encountered in our data. The model doesn't simply correct grammar but keeps the context and all the little nuances the speaker would imply. This was achieved by training the LLM on a massive amount of interview transcription corpus and feedback data to make it better placed in language patterns, plus common pitfalls

3) Evaluation of Competence

To conclude the competence of a user in communication, we take the difference between the original text and the one free of errors, with parameters such as number of corrections, type of grammatical mistakes made, and fluency in general. Such quantitative analysis unfolds a vast horizon in regards to user communication skills, along with understanding the areas for further improvement.

4) Enhanced Feedback Mechanisms

This helps the user to have feedback in multiple modes to help understand and learn better. They can listen to their corrected speech in different tones, formal and informal, through conversion from text to speech. User options may include listening to their speech in other languages to assist one in detecting errors and understanding better. Thus, such multimodal feedback would act by various learning preferences and aid in letting the user understand how much difference a tone or language will make in communication.

C. Multimodal Analysis: Assessing Confidence with CNN Integration

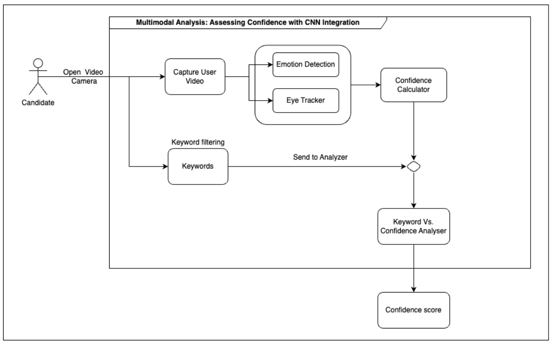

Fig 5: Multimodal analysis Module Architecture

In this module video recordings of candidates participating in mock interviews are collected. These recordings serve as the base dataset for training and evaluating CNN models. Each video is labeled with time stamps to facilitate detailed analysis of confidence levels at specific moments. Trained CNNs process the data using these models and provide instant feedback on the candidates' confidence levels.To ensure the accuracy and reliability of the data, preprocessing is performed using OpenCV (cv2). This includes tasks such as video frame extraction, normalisation, and upscaling to improve the robustness of the models. Facial landmarks and eye tracking data are also extracted using Dlib, providing essential elements for analysis. The core of the system is built using the TensorFlow ML Framework and Keras API. Some of the key aspects of this process are listed below.

1) Facial Landmark Recognition

Using Dlib and OpenCV, the system recognises and analyses facial landmarks to interpret expressions and emotions, which are critical indicators of confidence.

2) Eye Tracking

Implementing eye tracking algorithms to monitor gaze patterns and eye movements, providing insights into a candidate's attention and engagement levels.

3) Multimodal Integration

Combining facial landmarks, eye tracking data, and other features such as vocal tone and body language to provide a comprehensive assessment of confidence. CNN models synthesise this information to generate a holistic evaluation.

By integrating Python, CNN Deep Learning, OpenCV (cv2), TensorFlow ML Framework, Keras API, and Dlib, researchers are building this advanced system to assess confidence levels in mock interviews. Utilising these technologies, the project aims to provide IT graduates with a powerful tool to improve their interview performance and bridge the gap between academic preparation and career aspirations.

D. Personalised Skill Matching and Recommendation using RAG Model

The proposed approach leverages the RAG model to match job seekers' skills with relevant job opportunities and provide personalised recommendations for skill development.

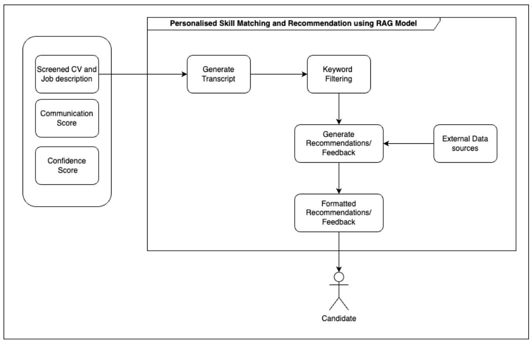

Fig 6: Multimodal analysis Architecture

1) Skill Extraction and Representation

The Retrieval-Augmented Generation (RAG) model can extract relevant skills from job seekers' resumes, cover letters, and other professional documents. It uses textual descriptions and structured data like skill taxonomies and ontologies for accurate skill representation. By incorporating contextual information such as industry, job role, and expertise level, the RAG model enhances skill identification and categorisation, ensuring alignment with specific job requirements and improving job matching.

2) Job Opportunity Analysis

Using the Retrieval-Augmented Generation (RAG) model, job descriptions and requirements can be analysed to identify desired skills and qualifications. These job opportunities are represented in a format similar to skill representations, utilising both textual descriptions and structured data for easy comparison and matching. Additionally, factors like company culture, location preferences, and career aspirations are incorporated to refine the matching process, ensuring a comprehensive and accurate alignment between job seekers and opportunities. This approach enhances the precision of job matches by considering both skill requirements and personal preferences.

3) Personalised Skill Matching and Recommendation

Develop a similarity metric to compare the skill representations of job seekers and job opportunities, incorporating contextual information such as industry, job role, and expertise level. Implement a ranking and recommendation system to suggest the most relevant job opportunities based on skill matches. Additionally, identify skill gaps and provide personalised recommendations for skill development, such as online courses, training programs, or mentorship opportunities, to enhance the job seeker's qualifications.

IV. Results and Discussion

A. Optimising Mock Interview Question Generation

To assess the precision of our resume and job description similarity scoring system, we employed a comprehensive evaluation framework involving a test dataset with annotated ground truth labels. The evaluation process begins by defining true positives (TP), false positives (FP), and false negatives (FN) within the context of similarity identification. True positives are instances where the system correctly identifies a high similarity between a resume and a job description, while false positives occur when the system incorrectly classifies a low similarity pair as high, and false negatives represent instances where high similarity is missed. The precision metric, calculated as TP / (TP + FP), provides a quantifiable measure of the system's accuracy in identifying relevant matches. Our evaluation involves running the system on a manually curated dataset, where we engaged five professional hirers who manually annotated a set of resumes and corresponding job descriptions by assigning similarity scores ranging from 0% to 100%, comparing the system's output against known similarity scores, and analyzing precision results to identify areas for improvement. The results obtained using the different models are as follows,

Table 1: Model Comparison for Job matching

Model	Mpnet	Roberta	Gpt3	Bert
Precision	0.68	0.76	0.52	0.84

The results of our evaluations indicate that that the 'Bert' model is highly effective for skills matching due to its superior precision.

The results demonstrate substantial advancements in question generation through the integration of skill clustering and augmented prompt techniques. The fine-tuning process, particular with the 10 epochs, yielded notable enhancements across various evaluation metrics. BLEU scores increased to 20.00, indicating better similarity between generated questions and human references. Rouge metrics showed considerable improvements, with Rouge1, Rouge2, and RougeL scores reaching 47.69, 26.43, and 44.15 respectively, reflecting enhanced precision and recall in question quality assessment.

Table 2: Hyper-parameters for fine-tuning instances

Model	Epochs	Batch Size	Warmup Steps	Weight Decay	Gradient accumulation steps	Learning rate	Save total limit	Fp16
T5-Small-FFT-V1	3	32	500	0.01	-	-	-	-
T5-Small-FFT-V2	10	16	1000	0.01	4	5e-5	2	True

Table 3: Evaluation metrics for the two instances

Model	BLEU	ROUGE 1	ROUGE 2	ROUGEL	ROUGELSum	METEOR	BertScore
T5-Small-FFT-V1	3	32	500	0.01	-	-	-
T5-Small-FFT-V2	10	16	1000	0.01	4	5e-5	2

These results underscore the effectiveness of integrating retrieval-based strategies with generative models, leveraging comprehensive knowledge retrieval from sources. The discussion highlights the broader implications of such advancements in educational settings, where tailored question generation can facilitate deeper comprehension and engagement.

B. Speech Empowerment Through Intelligent Verbal Analysis

The results from our pilot work suggest that using Whisper for the transcription, in combination with an optimised LLM for the grammar correction, will collectively augment the degree of accuracy and relevance in the feedback provided to the user. With this comes the gap analysis—quantitative differences between the original and corrected transcripts—giving an objective and precise measure that is quantifiable for communications proficiency and what is to be worked on. For example, users who frequently commit many grammatical mistakes in their oral text would then compare their original and corrected transcripts, quickly identifying the pattern to help target areas. Multimodal Feedback and User Engagement The most overwhelming is the fact that the feedback is of tones and languages, multiple, to bring out the rare dimension of our platform. People have reported being able to listen to their corrected speech in a formal tone, which helps them to prepare for professional settings, and the informal tone can be helpful for casual conversations. The ability to hear their speech in multiple languages has benefited non-native English speakers, who can get a better understanding of their mistakes and improve their language. This feature is not only supportive in recognising mistakes but also makes it easier for the users to understand context and the effective use of different styles of speech

C. Multimodal Analysis: Assessing Confidence with CNN Integration

Convolutional Neural Networks (CNNs) are an innovative way to evaluate and boost IT graduates' confidence that may be integrated into mock interview platforms. The system provides a comprehensive assessment of candidate confidence by leveraging multimedia data including facial expressions, eye movements and body language. This research began by collecting video recordings of candidates during mock interviews and meticulously pre-processing them using OpenCV (cv2) to ensure data robustness. Critical features such as facial landmarks and eye tracking data were extracted using Dlib and the core system was developed using TensorFlow and Keras to train and evaluate CNN models.

CNNs effectively detected facial landmarks and tracked eye movements, capturing key indicators of confidence. Compared to conventional approaches, the multimedia integration of body language, eye tracking, and facial emotions produced a more complete and accurate evaluation. Candidates may pinpoint and improve their areas of weakness with the use of the system's instant feedback, which helps them perform better during the interview. The researchers believe that this invention presents a powerful tool for IT graduates in the competitive job market.

D. *Personalised Skill Matching and Recommendation using RAG Model*

The proposed approach will be evaluated using real-world job seeker and job opportunity data. The evaluation metrics include precision, recall, and overall matching accuracy, as well as qualitative feedback from job seekers and employers. Additionally, the system's ability to provide actionable skill development recommendations will be assessed.

One of the key challenges in implementing this approach is the availability of high-quality training data for the RAG model. While there are publicly available datasets for job descriptions and resumes, they may not accurately represent the diverse range of skills and industries present in the job market. To address this challenge, techniques such as data augmentation, transfer learning, and active learning can be explored.Another challenge lies in the interpretability and transparency of the RAG model's decisions. As the model becomes more complex and incorporates various sources of information, it may become difficult to explain the reasoning behind its recommendations. Efforts will be made to develop interpretable models and provide clear explanations to users, ensuring trust and adoption of the system.

V. Conclusion

In conclusion, this research presents an advanced AI-powered mock interview platform designed to enhance employability for IT undergraduates. By leveraging Large Language Models (LLMs), Convolutional Neural Networks (CNNs), and Graph Neural Networks (GNNs), the platform offers personalised interview preparation through tailored questions, multimodal confidence analysis, and real-time communication feedback. Additionally, a Retrieval-Augmented Generation (RAG) model enables precise skill matching and provides targeted recommendations for skill development. This comprehensive approach addresses the dynamic challenges of the IT job market, equipping candidates with the necessary tools and confidence to succeed in securing desirable employment opportunities. Future directions for this research include expanding the platform's capabilities to support diverse industries, improving model interpretability, and enhancing data quality through advanced techniques like data augmentation and transfer learning. Additionally, integrating virtual reality (VR) for immersive interview simulations could further enhance candidate preparation and skill development.

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